

## PEMROGRAMAN MOBILE

### TUGAS API

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#### Link Github:

<https://github.com/rikafaulianarahmi/Mobile-Teori/tree/main/TugasRestApi>

#### ApiService.kt

```
1 package com.example.bookapp
2
3 import retrofit2.http.GET
4
5 interface ApiService {
6     @GET("books")
7     suspend fun getBooks(): ApiResponse
8 }
```

#### ApiResponse.kt

```
1 package com.example.bookapp
2
3 import
4     kotlinx.serialization.InternalSerializationApi
5     import kotlinx.serialization.Serializable
6
7 @OptIn(InternalSerializationApi::class)
8 @Serializable
9 data class ApiResponse(
10     val message: String,
11     val code: String,
12     val data: List<Book>
13 )
```

## RetrofitClient.kt

```
1 package com.example.bookapp
2
3 import
4   com.jakewharton.retrofit2.converter.kotlinx.serialization.asConverterFactory
5   import kotlinx.serialization.json.Json
6   import okhttp3.MediaType.Companion.toMediaType
7   import okhttp3.OkHttpClient
8   import retrofit2.Retrofit
9
10 object RetrofitClient {
11     private const val BASE_URL = "https://rika-
12     book-api.free.beeceptor.com/"
13
14     private val json = Json {
15         ignoreUnknownKeys = true
16     }
17
18     private val client =
19     OkHttpClient.Builder().build()
20
21     val apiService: ApiService by lazy {
22         Retrofit.Builder()
23             .baseUrl(BASE_URL)
24             .client(client)
25             .addConverterFactory(json.asConverterFactory("appli
26             cation/json".toMediaType()))
27             .build()
28             .create(ApiService::class.java)
29     }
30 }
```

## Book.kt

```
1 package com.example.bookapp
2
3 import
4   kotlinx.serialization.InternalSerializationApi
5   import kotlinx.serialization.Serializable
6
7 @OptIn(InternalSerializationApi::class)
8 @Serializable
9 data class Book(
```

|    |                     |
|----|---------------------|
| 9  | val id: Int,        |
| 10 | val title: String,  |
| 11 | val author: String, |
| 12 | val year: String,   |
| 13 | val genre: String   |
| 14 | )                   |

## MainActivity.kt

```

1 package com.example.bookapp
2
3 import android.os.Bundle
4 import android.widget.TextView
5 import androidx.appcompat.app.AppCompatActivity
6 import kotlinx.coroutines.CoroutineScope
7 import kotlinx.coroutines.Dispatchers
8 import kotlinx.coroutines.launch
9 import kotlinx.coroutines.withContext
10
11 class MainActivity : AppCompatActivity() {
12
13     private lateinit var tvStatus: TextView
14     private lateinit var tvBooks: TextView
15
16     override fun onCreate(savedInstanceState:
Bundle?) {
17         super.onCreate(savedInstanceState)
18         setContentView(R.layout.activity_main)
19
20         tvStatus = findViewById(R.id.tvStatus)
21         tvBooks = findViewById(R.id.tvBooks)
22
23         fetchBooks()
24     }
25
26     private fun fetchBooks() {
27         CoroutineScope(Dispatchers.IO).launch {
28             try {
29                 val response =
RetrofitClient.apiService.getBooks()
30                 withContext(Dispatchers.Main) {
31                     tvStatus.text = "Message:
${response.message} | Code: ${response.code}"
32                     val sb = StringBuilder()

```

```

33         response.data.forEachIndexed {
34             index, book ->
35                 sb.append("--- Book ${index
36                 + 1} ---\n")
37                 sb.append("Title   :
38                 ${book.title}\n")
39                 sb.append("Author  :
40                 ${book.author}\n")
41                 sb.append("Year    :
42                 ${book.year}\n")
43                 sb.append("Genre   :
44                 ${book.genre}\n\n")
45             }
46             tvBooks.text = sb.toString()
47         }
48     } catch (e: Exception) {
49         withContext(Dispatchers.Main) {
50             tvStatus.text = "Error:
51             ${e.message}"
52         }
53     }
54 }
55 }
56 }

```

### activity\_main.xml

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <ScrollView
3      xmlns:android="http://schemas.android.com/apk/res/a
4      ndroid"
5      android:layout_width="match_parent"
6      android:layout_height="match_parent"
7      android:background="#F5F5F5"
8      android:padding="16dp">
9      <LinearLayout
10         android:layout_width="match_parent"
11         android:layout_height="wrap_content"
12         android:orientation="vertical">
13         <TextView
14             android:layout_width="match_parent"
15             android:layout_height="wrap_content"
16             android:text="Book List"

```

```

17         android:textSize="22sp"
18         android:textStyle="bold"
19         android:textColor="#333333"
20         android:paddingBottom="8dp"/>
21
22     <TextView
23         android:id="@+id/tvStatus"
24         android:layout_width="match_parent"
25         android:layout_height="wrap_content"
26         android:text="Loading..."
27         android:textSize="13sp"
28         android:textColor="#888888"
29         android:paddingBottom="12dp"/>
30
31     <TextView
32         android:id="@+id/tvBooks"
33         android:layout_width="match_parent"
34         android:layout_height="wrap_content"
35         android:textSize="14sp"
36         android:textColor="#222222"
37         android:lineSpacingExtra="4dp"
38         android:fontFamily="monospace"/>
39
40 </LinearLayout>
41
42 </ScrollView>

```

### Output Program:

